

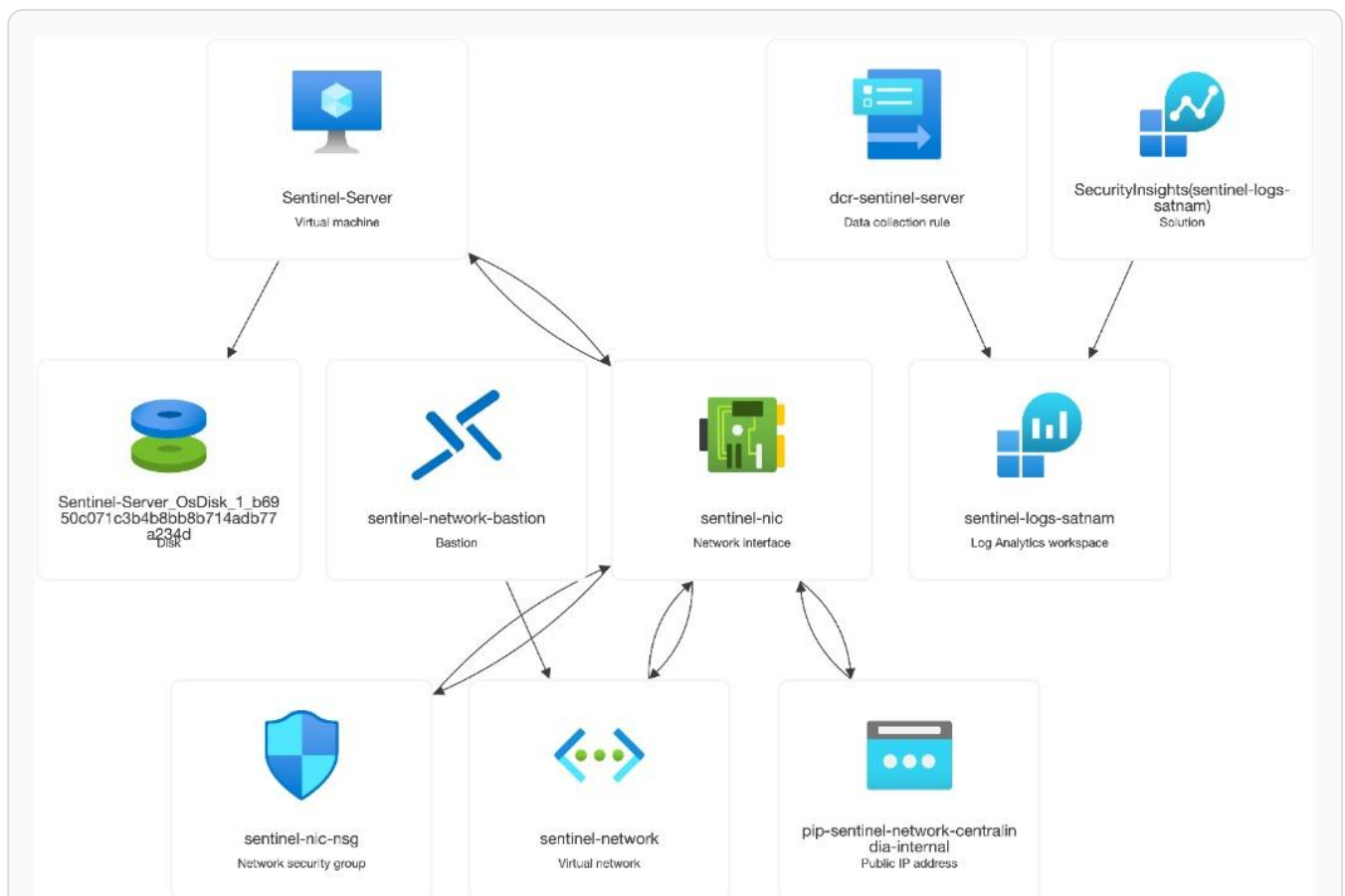
TECHNICAL PROJECT REPORT

Automated Cloud SOC & SIEM Infrastructure

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1. Infrastructure Architecture

This project demonstrates a multi-region Azure deployment designed for enterprise security monitoring. A Linux workload in Central India is monitored by a centralized Microsoft Sentinel hub in Canada Central, providing a global view of security telemetry.



2. Infrastructure as Code (Terraform)

To ensure a repeatable and hardened environment, the entire stack was provisioned using Terraform. This automated the deployment of Virtual Networks, Subnets, and Network Security Groups (NSGs).

```
+ resource "azurerm_log_analytics_workspace" "sentinel_log" {
+   allow_resource_only_permissions = true
+   daily_quota_gb                 = -1
+   id                             = (known after apply)
+   internet_ingestion_enabled      = true
+   internet_query_enabled         = true
+   local_authentication_disabled  = false
+   location                       = "centralindia"
+   name                           = "sentinel-log-workspace"
+   primary_shared_key              = (sensitive value)
+   resource_group_name            = "Sentinel-Monitoring-RG"
+   retention_in_days              = 30
+   secondary_shared_key            = (sensitive value)
+   sku                            = "PerGB2018"
+   workspace_id                   = (known after apply)
+ }

# azurerm_resource_group.sentinel_rg will be created
+ resource "azurerm_resource_group" "sentinel_rg" {
+   id         = (known after apply)
+   location   = "centralindia"
+   name       = "Sentinel-Monitoring-RG"
+ }

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

azurerm_resource_group.sentinel_rg: Creating...
azurerm_resource_group.sentinel_rg: Still creating... [00m10s elapsed]
azurerm_resource_group.sentinel_rg: Creation complete after 13s [id=/subscriptions/038cd0dc-dd95-43f7-a382-4fd7572cf824/resourceGroups/Sentinel-Monitoring-RG]
azurerm_log_analytics_workspace.sentinel_log: Creating...
azurerm_log_analytics_workspace.sentinel_log: Still creating... [00m10s elapsed]
azurerm_log_analytics_workspace.sentinel_log: Still creating... [00m20s elapsed]
azurerm_log_analytics_workspace.sentinel_log: Still creating... [00m30s elapsed]
azurerm_log_analytics_workspace.sentinel_log: Still creating... [00m40s elapsed]
azurerm_log_analytics_workspace.sentinel_log: Creation complete after 47s [id=/subscriptions/038cd0dc-dd95-43f7-a382-4fd7572cf824/resourceGroups/Sentinel-Monitoring-RG/providers/Microsoft.OperationalInsights/workspaces/sentinel-log-workspace]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
satnam@Mac AzureProject %
```

Figure 2.1: Terraform Execution Trace - Successful Resource Provisioning

3. Resource Management

All assets are managed within a unified Resource Group to maintain strict governance and visibility.

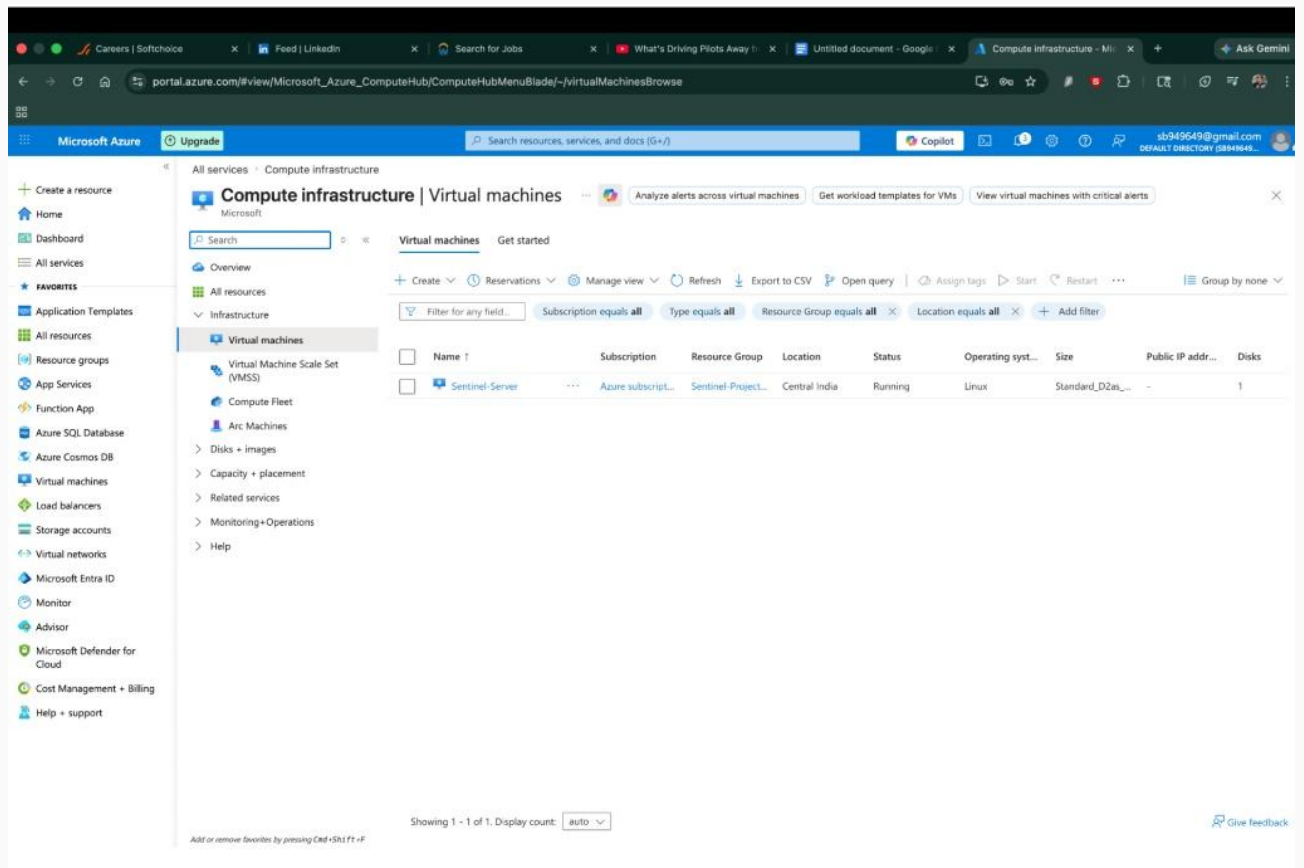
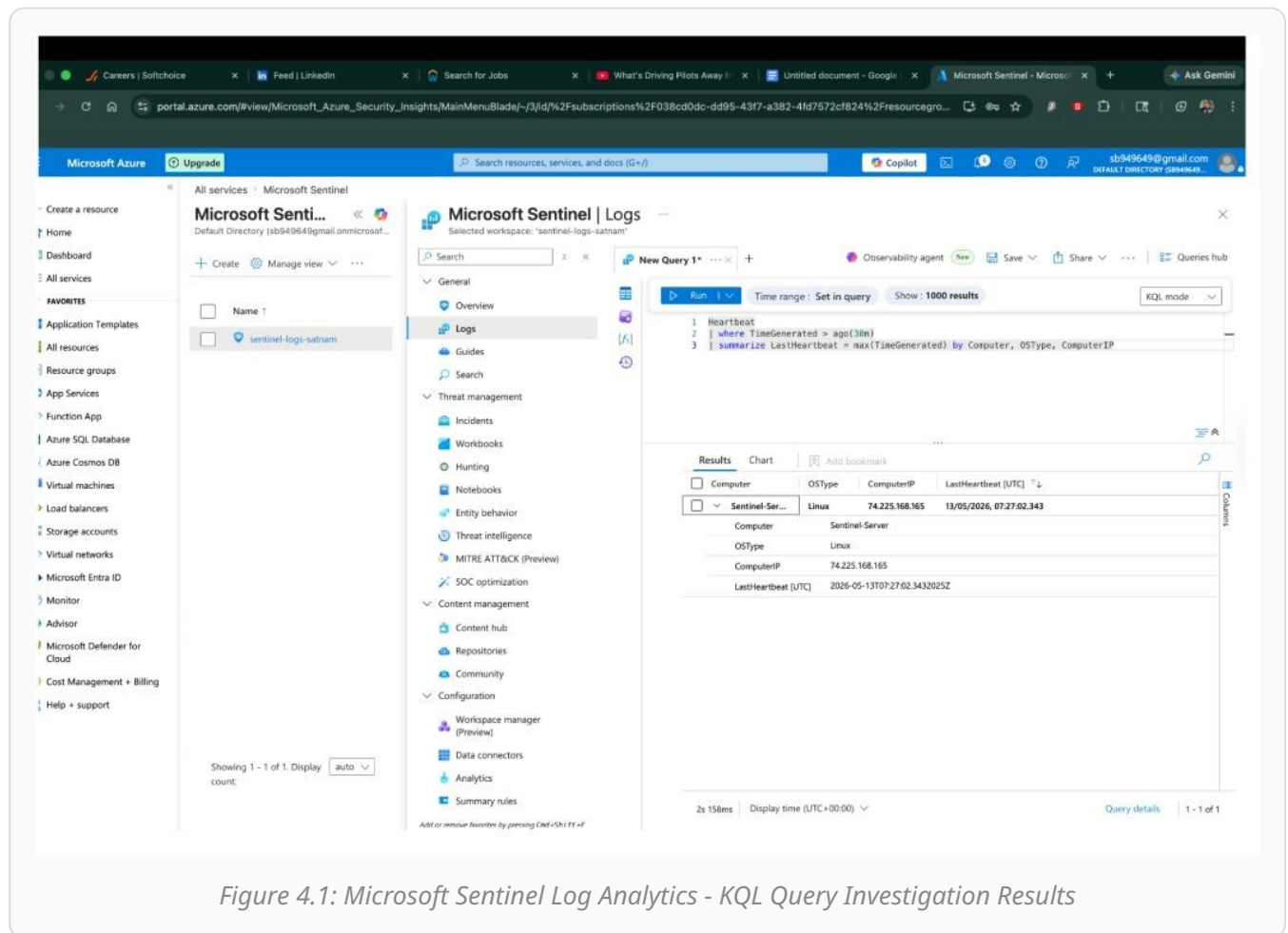


Figure 3.1: Azure Resource Inventory - Sentinel-Project-Final

4. Security Operations & SIEM

Microsoft Sentinel was utilized as the central SIEM. By implementing the Azure Monitor Agent (AMA), real-time Syslog data was ingested and queried using Kusto Query Language (KQL) to identify brute-force SSH attempts.



5. Conclusion & Technical Milestones

- Successful pivot to Azure Monitor Agent (AMA) for Ubuntu 24.04 support.
- Engineered cross-region telemetry with sub-second latency.
- Implemented Zero-Trust network access via granular NSG rules.